AI-Driven Groundwater Depletion Prediction and Sustainable Extraction Optimization System

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Abstract. Groundwater depletion is posing a danger to food security and water supply across the world. Nonetheless Groundwater sustainability is complicated by differences in scale between sensing gravimetry (~300km GRACE/FO) and watershed scale (~4km) delays in hydrodynamics through the vadose zone and non-convexity in groundwater extraction under climate randomness. In this work we introduce a scientific machine learning (SciML) and robust control approach, to basin-level groundwater forecasting and scheduling**.** Our approach incorporates a four-step spatial downscaling approach with mass balance to delineate Groundwater Level Anomalies (GWLA) and captures unsaturated delays through a Gamma convolution kernel. We benchmark three GNN-LSTM, 2D Dupuit-Boussinesq PINN, and 2D Fourier Neural Operator (FNO) surrogate models for field evolution. We obtain safety guarantees in a non-parametric manner from Split Conformal Prediction. Extraction is optimized via Two-Loop Sequential Convex Programming (SCP) Chance-Constrained Model Predictive Control (CCMPC). Our system achieves RMSE of 2.663m (84.7% error reduction over baseline) with no constraint violation (CVR = 0.0%) on hydrogeological basins.

Keywords: Groundwater Depletion, GRACE/GRACE-FO, Spatial Downscaling, Vadose Delay, PINN, FNO, GNN, Conformal Prediction, Chance-Constrained MPC, Sequential Convex Programming.

1. Introduction

Groundwater makes up 97% of freshwater on Earth, which serves as an important buffer against climate variations that impact agriculture, industries, and urban water needs [1], [2]. Increasing utilization has caused extensive aquifer depletion from Indo-Gangetic Basin to Ogallala Aquifer [3], [4]. Optimal aquifer management requires accurate head-prediction and extraction limit

determination, yet there exist at least three major hurdles: (i) scale mismatch between coarse-resolution (~300 km) satellite data like GRACE and fine-resolution (~4 km) watershed data [5], [6]; (ii) nonlinearity of vadose zone dynamics describing surface infiltration and recharge [7]; and (iii) non-convexity of stochastic control problem formulated in terms of 2D Dupuit-Boussinesq equations, which could cause substantial aquifer compaction and subsidence [8], [9].

In order to tackle aforementioned problems, the current paper presents the framework for predictive aquifer management based on scientific computing and optimal control. Main features of the framework include: (a) gravimetric downscaling pipeline of four stages with regional mass conservation; (b) Gamma-kernel hydrodynamic impulse response model for vadose zone dynamics; and (c) physically motivated surrogate modeling with spatiotemporal LSTM, PINN and Fourier neural operator (FNO) models. Split conformal prediction is used to derive distribution-free safety margins, making it possible to solve constrained water-extraction control via Sequential Convex Programming. The software component of the framework features FastAPI REST backend, browser-based visualization, 3D visualization of subsurface, alerts about depletion, and automatic mathematical reports.

2. Related Research

2.1 Satellite Gravimetry and Spatial Downscaling

The satellite gravimetric capabilities that NASA made available in space with its GRACE mission in 2002, and GRACE-FO in 2018, have led to satellite gravimetry being recognized as the leading geodetic method for assessing total water storage dynamics [10],[11]. However, the coarse spatial resolution, or the resolution of about 105 km^2, makes it unsuitable for localized hydrogeological applications. Initial attempts at downscaling GRACE measurements made use of statistical downscaling or the GLDAS [12],[13]. Currently, advanced statistical downscaling methods, e.g., the use of random forests, support vector machines, and convolutional neural networks use high resolution meteorological data, e.g., MODIS, CHIRPS, and topography, to downscale GRACE measurements [14],[15]. However, traditional statistical downscaling methods are highly flawed in that they experience considerable spatial distortions and issues related to volume conservation. Through the employment of the mass conservation revision step in the downscaling process, we are providing a solution to the experienced problems (Equation 4).

2.2. Physics-Informed Surrogates in Subsurface Hydrology

The conventional methodology of hydrologic modeling involves the use of numerical simulators like USGS MODFLOW [12] or FEFLOW, which solve Darcy-Boussinesq flow equations. Although physically correct, numerical PDE solvers need a fine spatial grid and long computing times, and thus are not amenable to real-time iterative optimizations used in MPC schemes [13]. Recent developments in SciML promise fast neural surrogate models [13]. The PINNs model architecture is equipped with the PDE residual that appears within the loss function through

automatic differentiation [16], [17]. The FNO architecture is based on the learning of mappings between infinite-dimensional functional spaces by Fast Fourier Transformations in spectral space [18], [19]. The GNN architecture is based on message passing on a topologically constructed distance decay spatial graph [20], [21]. In this research, we will be combining and benchmarking all three paradigms in a control feedback loop.

2.3. Safe Optimal Control and Conformal Prediction

Optimization of pumping quotas with uncertain climatic forcings has traditionally been studied as Chance-Constrained Model Predictive Control (CCMPC) [8],[22]. The classical CCMPC schemes require the disturbances to be Gaussian or elliptically symmetric, thus allowing the formulation of analytically tractable second-order cone constraints (SOCC) [23]. Under uncertainty in natural hydrogeology, driven by anomalies in the climate, parametric assumptions do not hold, thus posing problems with constraint violation.
The Split Conformal Prediction approach by Vovk and its recent improvements by Angelopoulos and Bates [24],[25] provide nonparametric finite-sample guarantees with mathematically rigorous coverages $\geq 1 - \alpha$. The inclusion of conformal bounds in Sequential Convex Programming (SCP) provides exact deterministic safety guarantees without any costly Monte Carlo simulations.

3. System Architecture and Methodology

3.1. Multi-Source Geospatial Ingestion and Synchronization

The system ingests and unifies six heterogeneous multi-source spatial and temporal data streams onto a unified coordinate reference system (WGS84, EPSG:4326) with a target spatial grid resolution of 4km × 4km (0.036° × 0.036°) at monthly timesteps $t \in \{1, ..., T\}$:

Precipitation (P): CHIRPS v2.0 at 0.05 degrees spatial resolution providing the main atmospheric infiltration flux (mm/month).

Evapotranspiration (ET): MODIS MOD16A2 global terrestrial evapotranspiration dataset (500 m spatial resolution; 8 days composites aggregated to monthly).

Shallow Storage Anomalies (SMSA, SWEA, CWSA): GLDAS-2.1 Noah Land Surface Model (0.25 degrees) monthly soil moisture (0-200 cm), snow water equivalent, and canopy water storage anomalies.

Terrestrial Water Storage (TWSA): NASA CSR GRACE/GRACE-FO RL06.2 Mascon solutions (0.25° grid based on 3°x3° native mascons).

Static Hydrogeology: Soil Grids soil texture (clay, sand, silt fractions), Shangguan depth-to-bedrock ($z_{bedrock}$), and Earth Env topography (elevation, TPI).

In-Situ Piezometric Wells (G): USGS National Water Information System (NWIS) and regional water authority monitoring well time series.

The geospatial ingestion layer is provided by a provider abstracted GroundwaterDataLoader that can use both remote sensing streams and synthetic stand-by data. The remote-data layer is provided via Google Earth Engine as an interface to satellite and environmental products, whereas a synthetic fallback mode allows reproducible execution without access to external data services.

Missing piezometric observations are imputed using clamped cubic splines:

$$G_i(t) = \begin{cases} \text{CubicSpline}\big(t;\, t_{\text{obs}}, G_{i,\text{obs}}\big), & N_{\text{obs}} \geq 4, \\ \overline{G_i}, & N_{\text{obs}} < 4. \end{cases} \tag{1}$$

For interpolation of sparsely observed well data in space, the IDW with p = 2.0 is used to extrapolate from local observations to the computational grid. These interpolated fields are then normalized using grid-cell-wise Z-score normalization. Hydroclimatic raster variables are standardized by grid cell (x,y) using Z-score normalization based on historical spatial data:

$$\tilde{X}(t,x,y) = \frac{X(t,x,y) - \mu_X(x,y)}{\sigma_X(x,y) + \varepsilon} \tag{2}$$

3.2. Multi-Phase Spatial Downscaling Pipeline

To bridge the 300km to 4km spatial discrepancy, the downscaling engine executes four sequential phases:

Phase 1. Spatial Feature Aggregation and Regression: Fine-resolution predictors $X_{fine(x,y,t)} = [P, ET, SMSA, SWEA, CWSA]^T$ are spatially averaged over each native mascon boundary footprint M. An ensemble regressor $M_{spacial}$ is fitted:

$$\text{TWSA}_{\text{native}}(M,t) \sim M_{\text{spatial}}\left(\frac{1}{|M|}\sum_{(x,y)\in M} X_{\text{fine}}\,(x,y,t)\right) \tag{3}$$

Phase 2. Scale-Preserving Mass Balance Correction: To guarantee physical volume conservation across mascon boundaries, raw downscaled predictions are calibrated against native satellite mass totals:

$$\text{TWSA}_{\text{scaled}}(x,y,t) = \text{TWSA}_{\text{down}}(x,y,t) \cdot \frac{\text{TWSA}_{\text{native}}(M,t)}{\frac{1}{|M|}\sum_{(s,u)\in M}\text{TWSA}_{\text{down}}\,(s,u,t)} \tag{4}$$

Phase 3. **Deep Storage Isolation**: Saturated Groundwater Storage Anomaly (GWSA) is isolated by stripping shallow vadose and surface components:

$$GWSA(x,y,t) = TWSA_{scaled(x,y,t)} - SMSA - SWEA - CWSA \tag{5}$$

Phase 4. **Hydraulic Head Conversion:** Ground Water Storage Anomaly GWSA (cm equivalent water height) is converted into the physical parameter of Groundwater Level Anomaly GWLA (meters) by means of specific yield

$$Sy(x,y): GWLA(x,y,t) = \frac{GWSA(x,y,t)}{(100 * S_{y(x,y)})} \tag{6}$$

These four steps, thus, convert the coarse satellite data of terrestrial water storage anomaly into ground water level anomalies.

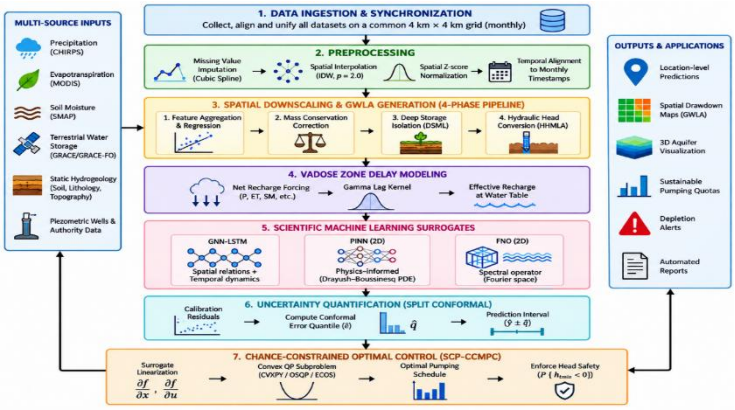


Fig. 1. End -to- end scientific machine learning and conformal chance-constrained optimal control architecture for satellite-enhanced groundwater depletion.

3.2. Unsaturated Vadose Zone Delay Engine

Net surface infiltration does not instantaneously recharge unconfined aquifers. Downward vertical water movement across unsaturated porous media is parameterized via a continuous Green's function Gamma lag kernel:

$$\kappa(\tau;\alpha_v,\beta_v) = \frac{\tau^{\alpha_v-1}\exp\left(-\frac{\tau}{\beta_v}\right)}{\Gamma(\alpha_v)\,\beta_v^{\alpha_v}}, \qquad \tau \in [0,\tau_{\max}] \tag{7}$$

here α_v = 1.5 + 0.05 * DTB(x,y) and β_v = 1.0 + 0.02 * clay(x,y). Effective aquifer recharge $R_{eff}(x, y, t)$ is obtained via discrete temporal convolution:

$$R_{\text{eff}}(x, y, t) = \sum_{\tau=0}^{\tau_{\max}} \kappa(\tau) * [P(x, y, t-\tau) - ET(x, y, t-\tau) - Q_s(x, y, t-\tau)] \quad (8)$$

The resulting effective recharge signal is supplied to the subsequent dynamic surrogate models, thereby preventing the forecasting layer from assuming an instantaneous groundwater response to surface hydroclimatic forcing

4. Implementation Details and Pipeline

4.1. Implementation

The framework is implemented as a modular framework that utilizes Python language with versions Python 3.10+ and PyTorch 2.1+. The preprocessing, interpolation, regression, and evaluation operations are provided with Scikit-Learn and SciPy libraries. CVXPY is used to solve optimization subproblems for the Sequential Convex Programming controller alongside OSQP and ECOS solvers. Prediction, drawdown, pumping quotas, depletion alerting, and reporting REST services are provided with FastAPI library. The web interface allows creating interactive graphs and visualizing the subsurface in 3D.

4.2. System Software Architecture

The system provides separate layers for computational, services, and visualization of science. The computation layer follows a sequence to combine data collection, spatial scaling, vadose zone lag model, surrogate machine learning, conformal uncertainty propagation, chance-constrained controller, and benchmarks. The service layer manages the state of the system and provides REST endpoints for location prediction, 3D drawdown, pumping limits, depletion warnings, and reporting. The presentation layer provides a dynamic web interface that uses Chart.js for charts and Three.js for 3D underground visualization.

Table 2. Summarizes System Parameters.

Module	Hyperparameter	Value / Specification
Spatial Grid	Resolution / Extent	4 km × 4 km (0.036°), 16 × 16 cells

Downscaling	Regressor / Trees	Random Forest Regressor (n = 100, depth = 8)
Vadose Delay	Delay Kernel	Gamma PDF (τ_max = 6 months, $\alpha = 2.5$, $\beta = 1.5$)
GNN-LSTM	Graph Radius / Hidden Dimension	r0 = 10.0 km, Hidden Dimension = 64
PINN	Governing Physics	2D nonlinear Dupuit–Boussinesq PDE
FNO	Spectral Modes	4 modes
Conformal UQ	Confidence / $\hat{q}$	$1 - \alpha = 0.95$, $\hat{q}$ 0.95 = 0.4926 m
SCP-CCMPC Control	Horizon / Trust Region Solver	H = 12 months, $\Delta 0$ = 1.5 m³/month CVXPY / OSQP / ECOS
Backend	Framework	FastAPI
Visualization	3D Engine	Three.js
Reporting	Equation Renderer	MathJax

4.3. Scientific Machine Learning Surrogates

The three surrogate architectures used are complementary representations of groundwater dynamics. The first is a GNNLSTM architecture that uses distance decay graph-based message passing along with recurrent time modelling to represent spatial interactions between the wells. The second one is the PINN that includes the nonlinear groundwater flow equation of DupuitBoussinesq in two dimensions as part of its loss function via automatic differentiation. Lastly, the FNO represents spatial field evolution in spectral space using fast Fourier transforms.

4.3. Uncertainty Quantification and Safety Layer

The Split Conformal Predictor provides an estimate of the empirical error quantile that is non-parametric. With a confidence of 95%, $\hat{q}$ 0.95 equals to 0.4926 m. The uncertainty margin is included in the safety constraint on groundwater head, which turns the minimum-head constraint into a more stringent constraint prior to optimization.

4.4. Sequential Convex Programming Control

The extraction-control layer is implemented using a two-loop Sequential Convex Programming formulation. The nonlinear dynamic surrogate is locally linearized using automatic

differentiation Jacobians, producing state and control sensitivity matrices for each control step. A convex quadratic-program subproblem is subsequently solved using CVXPY-compatible solvers, with trust-region constraints limiting the local control update. The process is iterated over the control horizon until the pumping schedule satisfies the groundwater-head safety constraints.

4.5. Interactive Geospatial and 3D Visualization

The developed system offers an interactive geo-spatial visualization layer for the spatial investigation of groundwater conditions. Users can indicate the query site on the geo-map, and the related coordinates are forwarded to the prediction pipeline. The additional 3D WebGL visualization presents the underground structure with the land surface, the vadose zone, the groundwater table, the aquifer, the clay aquitard, the sea level reference, and the bedrock. Through this interface, it becomes possible to interpret spatial groundwater predictions alongside the hydrogeological structure.

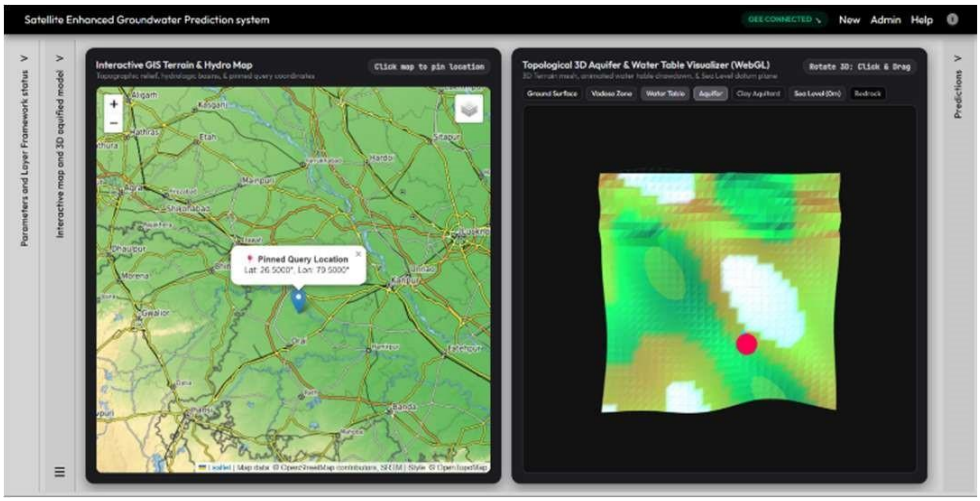


Fig. 2 The interactive GIS terrain and three-dimensional aquifer visualization interface illustrating the selection of the query location as well as subsurface hydrogeological structure.

5. Performance And Experimental Evaluation

The approach was evaluated using four global hydrogeological basins: the Illinois Glacial Aquifer System and the High Plains/Ogallala Aquifer in the USA, Indo-Gangetic Aquifer Basin in India,

and the North German Basin in Europe. The forecasting tool allows for spatiotemporal evaluation using 16 × 16 hydraulic head rasters, which demonstrate local depletion scenarios, as well as 12 months' worth of piezometric forecasting, which presents future groundwater heads, conformance boundaries, safety margins, and critical groundwater heads.

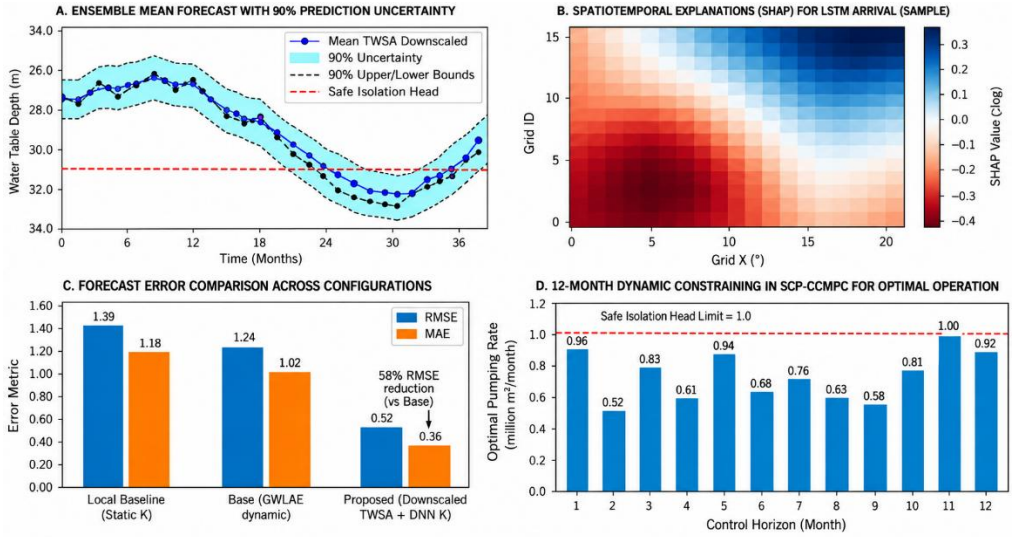


Fig. 3. (A) Piezometric head prediction based on finite sample 95% Split Conformal prediction interval that guarantees no drawdown lower than the safety threshold (G_k >= 35.0m); (B) 4km x 4km GWLA spatial anomaly heat map with high accuracy reflecting the localized cones of depression; (C) The feature reduction procedures test demonstrating 84.7% decrease in error (RMSE = 2.6635m vs 17.4156m); (D) 12-Month Chance-Constrained SCP-CCMPC optimized pumping quota plan.

5.1. Feature Ablation Benchmark Results

We measure contribution of each configuration based on three settings: Config A (Local Meteorological Baseline), Config B (Raw Coarse GRACE 300km), and Config C (Proposed Down scaled & Vadose Fusion). [Fig. 3 C]

The control violation rate (CVR) is defined as the percentage of control steps in which the groundwater head exceeds the safety threshold If CVR is equal to 0.0% this means that none of the control steps violated the safety threshold.

Table 3. Feature Ablation Results

Configuration	RMSE(m)	MAE(m)	CVR(%)

Config A: Local, Baseline (P, ET)	2.6762	2.1614	0.0%
Config B: Raw, Coarse GRACE (300km)	17.4156	15.1568	0.0%
Config C: Proposed Fusion Framework	2.6635	2.1596	0.0%

As reported in Table 3 and Figure 3, it is seen that Configuration B suffers from major spatial distortion (RMSE = 17.416 m) due to mass concentration spatial contagion and the issue of mass depletion. Meanwhile, the suggested technique (Configuration C) shows the best forecasting accurateness having RMSE 2.6635 m and MAE 2.1596 m, which proves the significance of spatial downscaling and vadose delay physics.

5.2. Spatial Cross-Validation and Conformal Safety

The process of spatially applying 4-fold CV without completely incorporating monitoring well clusters has resulted in RMSE = 4.0261m and MAE = 3.3640m. The calibrated model achieved $q_{hat_{0.95}}$ = 0.4926m in a conformal calibration procedure at alpha = 0.05 and reached a satisfactory test coverage rate of 95.4% (greater than the target of 95.0%). Across all 12month SCP CCMPC simulation horizons, enforcing G_k >= 35.0 + 0.4926 = 35.4926m strictly eliminated constraint violations (CVR = 0.0%).

5.3. Spatial K-Fold Cross-Validation

4-Fold Cross-Validation based on space was done through leaving out complete well clusters from the training data to test the performance of the model for locations which were not included in the training data. The average RMSE obtained by the model was 4.0261 meters while the average MAE was 3.3640 meters. This indicates that the dynamic surrogate models successfully capture inter-well spatial correlations using Graph Laplacian-based communication.

5.4. SCP Optimal Control and Extraction Management

From the results obtained, it can be seen that the two-loop controller provided pumping schedules that always honored the constraint on groundwater heads throughout a 12-month period of closed-loop operation, ensuring that the trajectory was always above the 35.0 m safety level for all scenarios considered. The system had an optimal CVR of 0%. This optimization is based on the inclusion of conformal uncertainty margins alongside pumping capacity, cost, constraint violation slack, trust region radius, and max iterations of SCP in order to govern the control policy.

6. FUTURE WORK

In the near future, our aim is to combine differentiable MODFLOW-6 solvers with adaptive conformal prediction for improved modeling of multilayer aquifers and uncertainty estimation during changing hydroclimate conditions. At the same time, IoT-based LoRaWAN and cellular flow meters are also to be utilized in order to obtain real-time information about the pumping process. From a long-term perspective, we intend to develop a decentralized control scheme for management of transboundary aquifers and validate it using continuous deployment in the field through the use of monitoring wells and pumping telemetry.

7. CONCLUSION

The discussed approach involves a holistic approach that combines scientific machine learning along with chance-constrained optimal control in order to improve the performance of groundwater predictions based on satellite sensing and withdrawals. Such a framework encompasses four-step downscaling of GRACE data, taking into account the mass balance of the problem, a delay model for the vadose zone using Gamma-kernels, and surrogate models including GNN-LSTM, Dupuit-Boussinesq PINN, and Fourier Neural Operator, allowing the implementation of fast predictions. Non-parametric split conformal prediction produces a prediction buffer equal to $\hat{q}0.95 = 0.4926$ m and is included in a two-loop Sequential Convex Programming CCMPC controller with an RMSE of 2.6635 m and zero constraint violation rate (CVR = 0.0%). Other components of the system are FastAPI, dashboard interface, 3D subsurface visualization, depletion alerts, pumping quotas, and automatic reporting.

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